

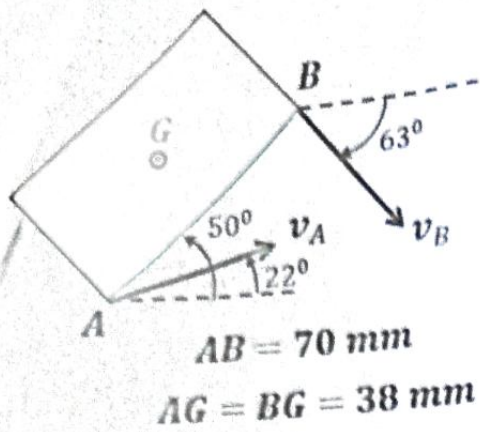


Figure 3

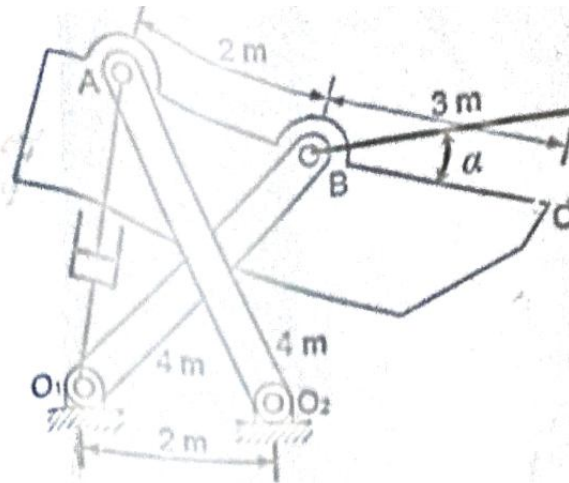


Figure 4

Question 6: In figure 5, the various dimensions are: $e = 5 \text{ cm}$, $CE = 50 \text{ cm}$, $AB = 20 \text{ cm}$, $BC = 25 \text{ cm}$. Find all ranges of values for AD and CD such that the entire chain is Grashof. Ignore condition for transition linkage. [15 marks]

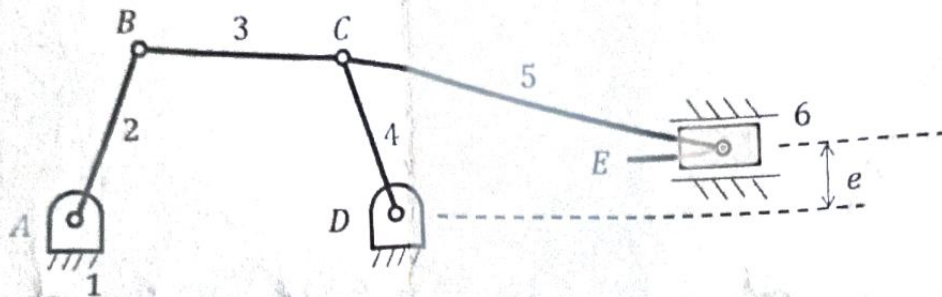


Figure 5

Question 7: Enumerate all combination of links possible in the case of an 8-link chain with 3 degrees of freedom. Draw atleast two enumerations for each scenario. [16 marks]

Question 8: Define the following terms with an example; (i) Kinematic inversion, (ii) Transmission angle, (iii) Dead Point, (iv) Toggle Position. [16 marks]

Question 9: Draw the kinematic diagram of the excavator in Figure 6 and compute the degrees of freedom. [18 marks]

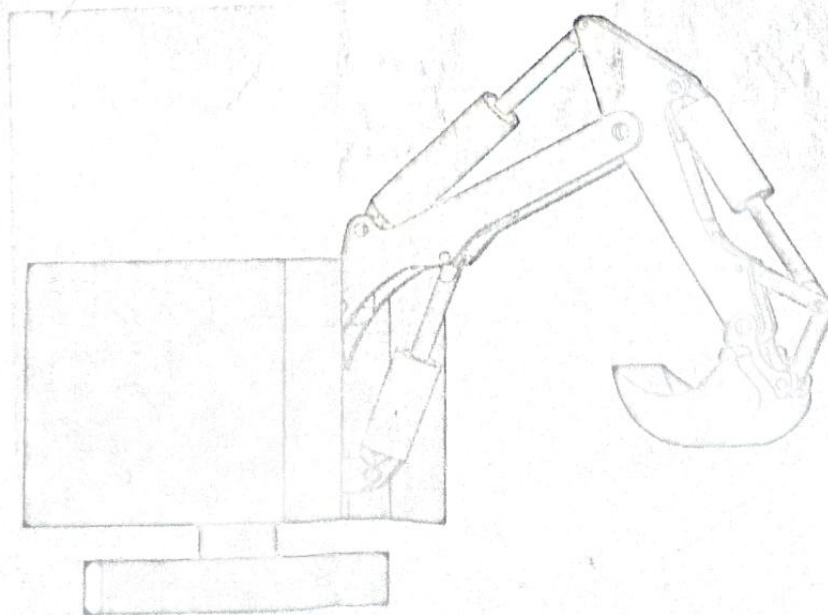


Figure 6